Revisiting Multi-Agent Social Reasoning for Conversational Power: Factor Graphs, Decision Models, and Context

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Abstract

Wu et al. (2025) predict which of two speakers in a six-turn meeting excerpt is the professor, with a factor graph over LLM-written reasons and rebuttals. We proposed to improve the graph with quote-checked arguments and Dung's grounded extension. On the 151 released snippets neither change helps, and no method we tried scores above the majority label. We then rebuilt the snippets from the ICSI Meeting Corpus, taking speaker turns from the corpus itself. On the 150 rebuilt snippets a System One decision model (Jev) identifies the professor in 71.3% of cases against a 56.7% majority baseline, and in 89.3% when it also sees 60 segments of meeting on each side. Their factor graph, rerun call by call with arguments from GPT-6.1 Terra and scores from Jev, reaches 72.0%, level with asking Terra directly (72.7%). A factor graph built from typed yes/no statements, with no generated arguments, matches the direct question at every context length.

1 Introduction

Wu et al. (2025) study professor-student exchanges of six turns from the ICSI Meeting Corpus (Janin et al., 2003) and take the professor to hold more power. They report 46.3% for six PhD judges, 49.0% for GPT-3.5 on the snippet alone, and 54.3% for a factor graph over LLM-written reasons and rebuttals solved with AD3 (Martins et al., 2011). In the label file they released, speaker A is the higher-power speaker in 83 of 151 rows, so always answering A scores 55.0%.

Our proposal kept their cached arguments and changed three things: a quote check on every argument, a rebuttal prompt that shows the rebuttal, and Dung's grounded extension (Dung, 1995) in place of their XOR program. Section 2 reports those experiments on the released snippets. Section 3 is reserved for the description of our data. Section 4 reports the same task on snippets rebuilt from the corpus, with the original factor graph, a decision model, zero-shot LLMs, and longer context.

Method	Correct	Acc.
Always A	83	55.0
XOR graph, cached GPT-3.5 scores	82	54.3
Grounded extension, as proposed	69	45.7
XOR graph with quote filter	64	42.4
Gemma 4 zero-shot	66	43.7
Jev, "is X the professor"	72	47.7
XOR graph, Jev scores	76	50.3
Preference grounded, cached scores	79	52.3
Preference grounded, Jev scores	71	47.0
Typed-statement graph (Jev)	77	51.0

Table 1: Accuracy on the 151 released snippets. Beating the 55.0% base rate at $p < 0.05$ needs 94 correct.

2 The released snippets

Table 1 lists the runs. Our reimplementation of the XOR graph as an integer program reproduces the published 54.3% exactly (82 of 151). The quote filter kept 33.8% of the 7,248 arguments and lowered accuracy.

The grounded extension is degenerate as we specified it. Every reason has one paired rebuttal and nothing attacks a rebuttal, so every rebuttal is IN and all 3,624 reasons are OUT. All 151 decisions fall to the tie-break. Removing weak rebuttals does not help, because we also made each reason for A and each reason for B attack each other, and under grounded semantics a mutual attack leaves both arguments undecided. A preference-based framework, where an attack succeeds only when the attacker scores at least as high as its target, removes both problems and gives a decision from IN counts on every snippet. It is well behaved and no more accurate (79 of 151 with cached scores).

Rescoring does not help. A logistic regression fit to the cached validity and grounding scores, evaluated leave-one-out, reaches 53.0 to 57.0%.

Rescoring all arguments with Jev, including the corrected rebuttal question on all 3,624 rebuttals, gives 50.3% under the XOR graph.

3 Data

This section is intentionally left incomplete in this draft. It will describe the rebuilt snippet set and how it relates to the released file. The construction procedure is summarized at the start of Section 4 and documented step by step, with code that rebuilds the data from its public sources, in the project repository.

4 Snippets rebuilt from the corpus

Construction. We located each released snippet in the original ICSI transcripts by exact match on its sequence of segment texts. All 151 snippets match exactly one place in the corpus. We then rebuilt each snippet from those corpus segments: speakers are the corpus participant IDs, consecutive segments by one participant form a turn, and the professor is the participant whose education level in the corpus metadata is Professor. One snippet contains a single speaker and is dropped, leaving 150, with the professor as A in 85 (a 56.7% base rate). For longer context we add 15 or 60 corpus segments on each side, with other participants shown as C, D, and so on. The released and rebuilt inputs differ in how turns are segmented and assigned to speakers; we defer discussion of that difference to Section 3.

Jev. Jev is a decision model that returns a probability for each typed yes/no question about a supplied state.[1] The direct question is "Is speaker X the professor?", asked for A and for B, and we predict the speaker with the higher probability.

The factor graph of Wu et al. We ran their method on the rebuilt snippets with arguments from current models, in two ways. In the first, GPT-6.1 Terra wrote the arguments call by call with their prompt templates: one independent request for each aspect summary, each list of three reasons, and each rebuttal (36 requests per snippet), with tools disabled and no access to labels. In the second, Claude Haiku wrote all arguments for a snippet in one pass. Jev scored both sets: validity and grounding of each argument, whether each rebuttal counters its reason, and similarity and contradiction of reason pairs. A separate Haiku instance also scored the Haiku arguments. Their rule file and weights (reason 20, rebuttal 18) do the inference, in the three variants of the paper.

[1] TypeSafe AI, `jev-1.14.0`, https://docs.typesafe.ai. The questions used on the rebuilt snippets were fixed before that data was built.

	Rel.	Rebuilt	±15	±60
More words	46.0	66.7	66.0	68.7
Longer turns		68.0	69.3	66.7
Jev, professor	48.0	71.3	84.7	89.3
Jev, more power	48.0	70.7	79.3	88.0
Typed-statement graph	51.3	70.0	80.7	90.7

Table 2: Accuracy (%) on the 150 snippets. "Rel." uses the released text and labels for the same snippets. The other columns use the rebuilt text at the same span and with 15 or 60 extra segments per side. Always answering A scores 56.7%. The blank cell was not run.

Grounded extension. We also ran the aggregator from our proposal on the same arguments. The framework has 48 arguments per snippet, 24 reasons and their 24 rebuttals. A reason has strength $P(\text{valid}) \cdot P(\text{grounded})$ and a rebuttal has strength $P(\text{counters its reason}) \cdot P(\text{grounded})$. In the *proposed* relation every rebuttal attacks its reason, and every reason for A and every reason for B attack each other. In the *preference* relation an attack succeeds only if the attacker is at least as strong as its target: a rebuttal defeats its reason when it is no weaker, and a reason defeats an opposing reason when it is strictly stronger. The grounded labelling marks an argument IN when all its attackers are OUT and OUT when some attacker is IN. The speaker with more IN reasons is predicted, with ties going to the larger sum of reason strengths and then to A.

Typed-statement factor graph. We also keep their graph and replace the generated text. For each of their four aspects we wrote three yes/no statements, such as "Does speaker X evaluate speaker Y's work?". Jev's probability for a statement about X is the weight of a reason for X, in the manner of Kassner et al. (2023), and the same statement about the other speaker is its rebuttal. One call with 24 questions replaces the generation and scoring calls for 48 arguments per snippet. Unlike the argumentation scoring studied by Sanayei et al. (2025), no model is asked to reason over the graph.

Results. Table 2 gives accuracy by input for the methods that need no generated arguments. On the rebuilt snippets at the same span, the direct

Args.	Scores	Dir.	Reas.	+Reb.	All	Gr.
Terra	Jev	72.7	66.7	72.0	72.0	70.0
Haiku	Jev	68.0	73.3	70.0	70.0	73.3
Haiku	Haiku	68.0	68.7	67.3	68.0	64.7

Table 3: Argument-based methods on the rebuilt snippets at the same span (accuracy, %). Rows give who wrote and who scored the arguments. "Dir." is the argument-writing model asked directly. "Reas.", "+Reb." and "All" are the three variants of the factor graph of Wu et al. (2025). "Gr." is the grounded extension with the preference relation. For comparison, the direct Jev question scores 71.3% and Claude Sonnet asked directly scores 76.0%.

Jev question identifies the professor in 107 of 150 cases. Against always answering A this is 34 wins and 12 losses (exact McNemar $p = 0.0016$), and a bootstrap that resamples whole meetings gives a 95% interval of 6.7 to 22.0 points for the difference. The same question on the released text scores 72 of 150, and the paired contrast between the two inputs is 52 wins to 17 losses ($p < 0.0001$). Context helps further: ± 15 segments against the same span is 30 wins to 10 losses ($p = 0.002$). The word-count rules reach 66.7% and 68.0%, but their paired tests against the majority label are not significant ($p = 0.10$ and $p = 0.053$).

Table 3 gives the argument-based methods. With Terra's arguments and Jev's scores the full factor graph of Wu et al. (2025) reaches 72.0%. That is level with Terra asked directly (72.7%, 14 wins to 15 losses) and with the direct Jev question (14 to 13). With Haiku's arguments it reaches 70.0% under Jev's scores and 68.0% under Haiku's, against 68.0% for Haiku asked directly. The parts of the graph do not behave consistently across the two sets of arguments. Rebuttals add 5.3 points with Terra's arguments (11 wins to 3 losses, $p = 0.057$) and cost 3.3 points with Haiku's under the same scorer, and adding the relations changes accuracy in neither Jev-scored run. A plain sum of reason scores with no graph equals the reasons-only graph in all three rows. The best single result at this span is a stronger model asked directly (Sonnet, 76.0%), which is not distinguishable from Jev (27 wins to 20 losses, $p = 0.38$) or from Terra (15 to 10, $p = 0.42$).

The grounded extension behaves on the rebuilt snippets as it did on the released ones. Under the proposed relation no reason is ever IN in any run, so all 150 decisions are made by the tie-break and the accuracy is that of the sum of reason strengths. The preference relation gives a real decision: with Jev scores every snippet is decided by IN counts, and accuracy is 70.0% with Terra's arguments and 73.3% with Haiku's, the latter level with the direct Jev question (13 wins to 10 losses, $p = 0.68$). With the coarser Haiku scores it reaches 64.7%. We also searched 288 grounded policies over the Haiku arguments and Jev scores, varying the strength definitions, the rule for when a rebuttal attacks (preference or a threshold on Jev's judgment that it counters its reason), the cross-side rule, and the decision rule. Choosing the best policy on a development half of the meetings and scoring it on the other half, over 500 random splits, gives 69.7% on average against 71.5% for the direct Jev question on the same halves, so no policy in this family improves on asking directly.

The typed-statement graph tracks the direct Jev question at every input (70.0% against 71.3% at the same span, 21 wins to 23 losses, $p = 0.88$). It is a cheaper way to run the method and exposes which statements drive a decision, and we do not claim it is more accurate than asking directly.

Jev's probabilities are informative on the rebuilt snippets. Its Brier score is 0.189 at the same span and 0.095 at ± 60, against 0.246 for the base rate, and its most confident fifth of answers is 93.3% correct at the same span.

5 Limitations

The order of work was exploratory. The rebuilt set was constructed after many runs on the released file, and every number in Section 2 was computed on test data with no development split. The questions used on the rebuilt snippets were written before that data existed, and each run was made once.

Neither rerun of the factor graph uses the original model. The Terra run follows the pipeline call by call, but the text comes from GPT-6.1 Terra and the scores from Jev, because the command-line interface we used does not expose token probabilities. The Haiku run wrote each snippet's arguments in one pass, so reasons were written with the conversation in view, and its Haiku scores are stated probabilities with about 15 distinct values. In both, each rebuttal is judged against its own reason and grounded on its own text.

The rebuilt set has four professors, and one of them is the professor in 110 of the 150 snippets. Jev's accuracy for that professor is 69.1% at the

same span, and 73.3 to 100% for the others, so the result is not carried by one person, but the meeting-level bootstrap is the more honest interval. Most of the non-professors hold a PhD, so "student" follows Wu et al. (2025) and not the metadata. With 60 extra segments a model can use cues that a six-turn excerpt lacks, including how others address the two speakers.

We did not rerun GPT-3.5, Gemma, or a human study on the rebuilt snippets. Jev is a proprietary model in early access whose training and calibration claims are not peer reviewed. The 40-item human audit from our proposal was not completed.

Earlier work predicts power from word choice and thread structure with supervised models (Danescu-Niculescu-Mizil et al., 2012; Prabhakaran and Rambow, 2014), and Ziems et al. (2024) find that LLMs often trail such models. Our numbers say only that the role is recoverable from these excerpts, and they are not a comparison with that line of work.

6 Reproducibility

The repository contains the rebuilt snippets with corpus offsets, every model answer, and all code.[2] One command downloads the five external inputs (the released `meeting.csv`, argument cache and rule file, and the two ICSI annotation archives), verifies each by SHA-256, regenerates the snippet set byte for byte, and recomputes every number in this report from the stored model answers without calling a model. The model answers themselves come from hosted models and cannot be regenerated exactly; the scripts and instructions used to collect them are included. The ICSI corpus is distributed under CC BY 4.0.

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[2] https://github.com/muttaqiahmadalladin/cs-577